

FAIRPORT, MI, USA

WMO: 997792

Lat: 45.6187N

Lon: 86.6596W

Elev: 610

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.3	(c) -0.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 34.0	(k) 14.9	(l) 31.0	(m) 17.8	(n) 13.9	(o) 300	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.0	(c) 76.4	(d) N/A	(e) 74.7	(f) N/A	(g) 72.8	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.3	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 28.8	(b) 25.9	(c) 23.5	(e) -7.8	(f) 80.8	(g) 7.8	(h) 3.3	(i) -13.4	(j) 83.2	(k) -18.0	(l) 85.1	(m) -22.4	(n) 87.0	(o) -28.1	(p) 89.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	18.7	18.7	27.7	38.3	48.9	58.6	65.7	66.1	59.7	49.0	37.6	26.7		
	DBStd	18.25	10.33	9.88	9.36	5.92	5.67	5.40	5.14	4.25	5.90	6.64	7.95	8.27		
	HDD50	4185	970	876	692	350	89	4	0	0	4	101	375	724		
	HDD65	8165	1435	1296	1157	800	501	201	55	37	174	497	823	1189		
	CDD50	1665	0	0	0	1	53	261	485	499	294	70	2	0		
	CDD65	168	0	0	0	0	0	8	75	71	14	0	0	0		
	CDH74	212	0	0	0	0	1	8	125	75	3	0	0	0		
	CDH80	6	0	0	0	0	0	0	6	0	0	0	0	0		
(14) Wind	WSAvg	10.6	12.2	10.7	9.9	10.3	9.6	8.4	8.8	9.3	10.5	12.2	13.6	12.0		
(15) Precipitation	PrecAvg	29.8	1.8	1.3	1.9	2.4	3.0	2.8	2.8	3.5	3.4	2.6	2.2	2.2		
	PrecMax	38.8	4.7	4.0	5.9	4.0	7.4	7.2	5.6	6.6	7.1	5.3	5.6	5.3		
	PrecMin	21.5	0.2	0.0	0.1	0.6	0.9	0.4	0.7	0.2	0.9	0.8	0.5	0.1		
	PrecStd	4.6	1.1	0.9	1.4	0.9	1.4	1.6	1.2	1.6	1.5	1.2	1.3	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.0	38.7	51.7	55.8	69.2	75.4	80.2	78.1	73.3	65.9	54.6	46.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	37.0	35.5	45.5	51.9	63.7	71.1	77.2	76.1	71.1	63.1	52.2	42.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	35.2	34.1	42.5	49.1	59.9	68.4	75.1	74.6	69.7	61.3	50.2	40.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	33.2	32.5	38.8	46.5	56.9	66.3	73.3	72.8	68.3	59.4	48.3	38.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.6	12.1	10.5	11.7	10.7	10.1	10.0	10.1	9.0	8.4	9.2		
		MCDBR	9.4	10.2	11.6	14.2	15.4	12.8	11.0	10.6	10.2	8.9	7.9	9.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.282	0.313	0.355	0.386	0.393	0.389	0.387	0.369	0.335	0.303	0.274		
	taud		2.492	2.416	2.401	2.373	2.355	2.386	2.412	2.412	2.457	2.523	2.536	2.516		
	Ebn at Noon		272	288	292	287	280	277	277	273	269	265	256	258		
	Edh at Noon		21	27	32	36	38	37	36	35	30	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		419	725	1126	1460	1783	1918	1972	1669	1269	767	434	323		
	RadStd		37	85	104	136	141	152	83	111	92	83	44	30		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page